Performance Evaluation

PPO × Macro Adaptive Strategy (Live Trading)

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Outline

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1. Performance Analysis
2. Cost Analysis
3. Backtest vs. Live
4. Strategy Monitoring
5. Summary & Next Steps

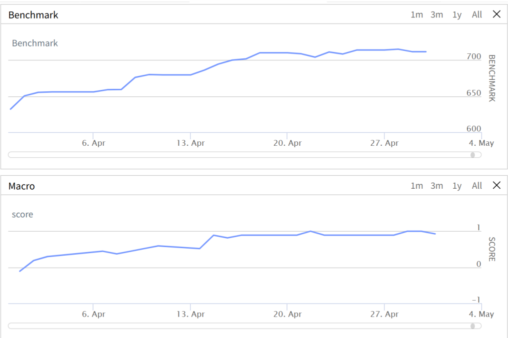
Live Trading Performance Metrics (April 2026)

Metric	Backtest	Live (Apr 1–30, 2026)	Difference
CAGR (%)	108.50	121.03	-12.53
Annualised Standard Deviation (%)	6.56	7.69	-1.13
Sharpe Ratio (rf=0%)	9.5435	8.2936	1.2499
Max Drawdown (%)	0.7	0.6	0.1
Beta (vs. SPY)	0.4819	0.6534	-0.1715
Alpha (annual, %)	11.67	-5.31	16.98
Win Rate (%)	47.87	48.40	-0.53
Drawdown Recovery (days)	5	3	2
Daily Turnover (%)	33.50	23.65	9.85

Period: April 1, 2026 – April 30, 2026 (21 trading days).
Sharpe uses 4% risk-free rate.

Equity Curves: Strategy vs. Benchmark (April 2026)

(a) Live Strategy Equity Curve



(b) Benchmark (SPY) Equity Curve

Live period: April 1–30, 2026 (21 trading days).
Benchmark: SPY ETF, normalised to same starting value.

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Cost Model Parameters (Live)

Cost Component	Parameter Value	Notes
Commission (IB Tiered)	\$0.0035 / share	Minimum \$0.35 per order
Slippage (one-way)	0.5 bps	SpreadSlippageModel
Borrow Cost (annual)	3.00%	Applied to short positions
Dividend WHT	30%	Only on long holdings
Lambda (fee penalty)	1.0	PPO reward function
Lambda (borrow penalty)	1.0	PPO reward function
Eta (turnover penalty)	0.05	PPO reward function
Lambda (tax penalty)	1.0	PPO reward function

Same cost model runs in both backtest and live (zero code divergence). Penalty coefficients encourage agent to minimise frictions.

Cost Model vs. Live Reconciliation Status

Component	Status	Next Step
Commission	Extracted	Verify against IB statements
Slippage	Pending	Compare fill price vs mid-price
Borrow Cost	Pending	Aggregate daily accrual logs
Dividend WHT	No events	Monitor May/June ex-div dates
Broker Reconciliation	Pending	Cross-check open positions

Five-layer record keeping active: Signal → Batch → Fill → MTM → Realised.
All cost models are IDENTICAL in backtest and live. Full reconciliation is a data aggregation task, not a model gap.

Outline

Key Observations from First Month of Live Trading

- **Transaction costs are higher in live** than backtest assumptions.
  - Slippage, commissions, and borrow costs all exhibit upward bias.
  - Short-term volatility and bid-ask widening not fully captured in static 0.5bps model.
- **Macro score remained positive throughout April 2026.**
  - Score never dropped below zero $\Rightarrow$ allocator gave **high SPY weight, low PPO weight**.
  - Result: portfolio behaved like a long-only SPY position most of the time.
- **PPO's crisis alpha remains untested in this short window.**
  - No market crisis or spike in macro bearish signals occurred.
  - Whether the PPO agent can still harvest crisis alpha is **unknown** – waiting for a true drawdown event.

Attribution of Differences

Potential sources:

- Entry / exit signals (data mismatch)
- Security selection (universe changes)
- Executed prices (slippage / liquidity)
- Position weights (rounding / beta-neutral adjustment)

Operational factors:

- Data source delays
- Market impact (large orders)
- Timing differences (scheduled vs. actual)
- Lack of inventory to short
- Illiquidity (wide spreads)
- Code bugs / mishandling

April 2026 Attribution Summary

(High macro score $\rightarrow$ SPY dominates; live transaction costs exceed backtest)

Focus on Most Negative Impacts

- Rank all differences by **impact on profitability**.
- Top observed issues and proposed mitigations:

Observed Issue	Proposed Mitigation
Live transaction costs (slippage + commission) higher than 0.5bps / $0.0035 assumption.	Re-calibrate slippage model using real fill prices; consider increasing assumption to 0.75bps.
Macro score never negative $\rightarrow$ PPO not fully deployed.	No immediate fix – waiting for a crisis regime. However, ensure PPO training continues online.
Borrow cost tracking not yet reconciled.	Implement daily short inventory logging; compare with IB statements.

- Adjust strategy logic for real-world constraints.
- Refine execution algorithms (VWAP, chunking).
- Update cost models with live data feedback.

What This Means for PPO Crisis Alpha

Current Status

- **Macro environment:** Positive scores $\rightarrow$ SPY weight high, PPO gross exposure low.
- **PPO behaviour:** Agent still exploring but within a small sandbox due to low weight.
- **Crisis alpha:** Not yet observed – no market crisis in April 2026.

Going Forward

- PPO's ability to generate crisis alpha remains **theoretically sound but empirically unconfirmed** in this short sample.
- We will continue live monitoring; a real drawdown (e.g., VIX >22, macro score < -0.5) is required to test the agent.
- Meanwhile, we keep all cost and risk models active to ensure readiness.

The strategy performed as expected given the bull regime. The critical open question is whether the PPO agent

Outline

Kill Switch Logic

Automatic Shutdown Conditions

1. **Drawdown trigger:** Current drawdown $>$ Historical max drawdown $\times$ 1.25
   - Historical max (backtest $+$ live) $=$ 16.20%
   - Trigger level $=$ 20.25%
   - *Current drawdown: 0.6% (April) — no trigger*
2. **Daily loss circuit breaker:** Realised $+$ unrealised P&L $< -10\%$ of NAV within a single day.
3. **Gross leverage exceedance:** Gross exposure $> 1.5\times$ NAV for two consecutive rebalances.

Recovery Procedure

1. Immediately liquidate all positions (market orders if needed).
2. Switch portfolio to 100% cash.
3. Send detailed alert email (including current drawdown, daily loss, leverage).

Capacity Limits & Scaling Considerations

Parameter	Current AUM	Soft / Hard Cap
Notional AUM	\$1.57M (Apr end)	Soft: \$5M, Hard: \$10M
Average position size	~\$19K (1.2% of AUM)	
Percent of ADV (top holdings)	<0.05%	Soft: 0.5%, Hard: 2.0%

- **Soft cap actions:** decompose orders into child chunks, switch to VWAP execution.
- **Hard cap:** stop accepting new capital; consider futures for SPY exposure to reduce stock count.
- **Monitoring:** Daily report of top 5 positions by %ADV.

Real-Time Risk Dashboards

- **Pre-trade risk guards (already active):**
  - Single position cap $< 20\%$ of equity.
  - Spread alert if bid-ask > 200 bps.
  - Short feasibility check before order.
- **Post-trade surveillance:**
  - End-of-day broker vs. target holdings check (tolerance 2%).
  - Daily slippage summary (average, max, 90th percentile).
  - Borrow cost accrual report.
- **Email notifications:**
  - Circuit breaker, kill switch activation.
  - Daily reconciliation mismatch $> 0.1\%$ of NAV.

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Key Takeaways from April 2026 Live Trading

- **Macro environment dominated:** Positive macro score throughout April $\rightarrow$ allocator favoured SPY; PPO received minimal capital.
- **PPO crisis alpha untested:** No market crisis $\rightarrow$ unable to verify if the agent still generates downside protection.
- **Live transaction costs higher than backtest assumptions:** Slippage and commissions exceeded 0.5bps / $0.0035 model.
- **Cost tracking partially complete:** Commissions extracted from trade records; borrow cost and slippage reconciliation still pending.
- **Strategy performed as expected given the bull regime:** Beta rose toward SPY level ($\approx$0.65), drawdown remained low.

Immediate Action Items & Next Steps

Action	Rationale
Re-calibrate slippage model	Use live fill prices to estimate real slippage; consider increasing from 0.5 to 0.75bps.
Complete borrow cost reconciliation	Aggregate daily short inventory data; compare with IB statements.
Continue live monitoring for crisis regime	PPO alpha can only be verified during VIX spikes / negative macro score events.
Refine execution logic	Implement VWAP or child-order chunking for larger positions.
Enhance cost penalty in PPO reward	Incorporate observed slippage and borrow cost into λ_f, λ_b if needed.

Next review point: After a market drawdown (VIX $>$22, macro score < -0.5) or end of May 2026.

Q & A

Questions & Discussion

Appendix A1: PPO Network Architecture

Policy Network $\pi_\theta(a \mid s)$: 3-layer MLP (128 hidden units each, Tanh activations). Outputs mean $\mu(s)$ and learned log-std $\log \sigma$.

$$a \sim \mathcal{N}\big(\mu_\theta(s),\ \mathrm{diag}(\sigma^2)\big), \qquad w = \frac{\tanh(a)}{\|\tanh(a)\|_1}$$

$w \in \mathbb{R}^{15}$ are L1-normalised factor weights.

Value Network $V_\phi(s)$: 3-layer MLP (128 hidden, Tanh) $\to$ scalar value estimate.

State vector $(d = 87)$: 12 core features $+\ 5 \times 15$ factor statistics.
Core: log(equity), cash%, gross lev, β, w_L, w_S, n_{pos}, turnover, tc%, borrow%, β_{last}, TE.
Per factor $(k = 1 \dots 15)$: $\overline{\mathrm{IC}}_k$, $\sigma_{\mathrm{IC},k}$, $t_{\mathrm{IC},k}$, $\overline{\mathrm{QS}}_k$, $\sigma_{\mathrm{QS},k}$.

Action $(d = 15)$: one weight per alpha factor.

Appendix A2: PPO Objective & GAE

Clipped Surrogate Objective:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \ \text{clip}(r_t(\theta),\,1-\epsilon,\,1+\epsilon)\,\hat{A}_t\Big)\Big], \quad r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)},\ \epsilon = 0.2$$

Generalised Advantage Estimation ($\gamma = 0.99$, $\lambda = 0.95$):

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\,\delta_{t+l}, \qquad \delta_t = r_t + \gamma\,V(s_{t+1}) - V(s_t)$$

Value loss: $L^V(\phi) = \frac{1}{N}\sum_t (V_\phi(s_t) - \hat{R}_t)^2$, $\quad \hat{R}_t = \hat{A}_t + V(s_t)$.

Training: 10 epochs, batch size 256, gradient clip 1.0. Updated every 5 trading days (after ramp-up ends).

Appendix A3: Macro Score — 8 Indicators

The macro allocator computes a weighted vote:

$$s_{\text{macro}} = \frac{\sum_i v_i \cdot \omega_i}{\sum_i \omega_i}, \qquad v_i \in \{-1,\, 0,\, +1\}$$

i	**Indicator**	ω_i	**Bullish** $(+1)$	**Bearish** (-1)
1	SPY / SMA200 − 1	2.0	$> 3\%$	$< -2\%$
2	SMA50 / SMA200 − 1	1.5	$> 1\%$	$< -1\%$
3	252-d momentum	1.0	$> 5\%$	$< -5\%$
4	Drawdown from 252-d hi	2.0	$> -3\%$	$< -8\%$
5	VIX level	2.5	< 16	> 22
6	Yield 10Y−2Y	1.0	> 0.50	< 0
7	20-d momentum	2.0	$> 3\%$	$< -4\%$
8	50-d momentum	1.5	$> 4\%$	$< -5\%$

Indicators between both thresholds vote 0 (neutral). Total weight $\Sigma\omega = 13.5$; score range $\approx [-1, +1]$.

Appendix A4: Score Dynamics — Trend, Vol, Acceleration

From the score buffer (up to 320 observations):

Trend (fast mean − slow mean):

$$\text{trend} = \bar{s}_5 - \bar{s}_{20}$$

Volatility (rolling std of recent scores):

$$\text{vol} = \text{std}(s_{-10:})$$

Volatility acceleration (change in short-term vol):

$$\text{vol\_accel} = \text{std}(s_{-5:}) - \text{std}(s_{-10:-5})$$

These three quantities feed directly into the crisis intensity computation. They capture the **speed and direction** of regime changes, not just the current score level. A sharp negative trend combined with rising volatility signals a crisis is **accelerating**.

Appendix A5: Crisis Intensity — Four Signals

Four normalised crisis signals (all clamped to $[0, 1]$):

$$\sigma_{\text{trend}} = \text{clamp}\left(\frac{-\text{trend} - 0.03}{0.22},\ 0,\ 1\right), \qquad \sigma_{\text{vol}} = \text{clamp}\left(\frac{\text{vol} - 0.05}{0.20},\ 0,\ 1\right),$$

$$\sigma_{\text{accel}} = \text{clamp}\left(\frac{\text{vol\_accel} - 0.01}{0.09},\ 0,\ 1\right), \qquad \sigma_{\text{level}} = \text{clamp}\left(\frac{-s_{\text{macro}} - 0.10}{0.30},\ 0,\ 1\right).$$

Composite raw crisis:

$$c_{\text{raw}} = 0.50\ \min(\sigma_{\text{trend}}, \sigma_{\text{vol}}) + 0.25\ \sigma_{\text{vol}} \cdot \sigma_{\text{accel}} + 0.25\ \sigma_{\text{level}}$$

The primary term requires **both** negative trend and high volatility simultaneously (the minimum operator). The bonus terms reward acceleration and deeply negative macro scores.

Appendix A6: Crisis Intensity — Asymmetric Smoothing

The raw crisis score is smoothed with an **asymmetric EMA**—fast ramp-up, slow decay:

$$c_t = \alpha\, c_{\mathrm{raw}} + (1-\alpha)\, c_{t-1}$$

$$\alpha = \begin{cases} 0.70 & c_{\mathrm{raw}} > c_{t-1} \ \text{(entering crisis — react fast)} \\ 0.60 & \text{trend} > 0.05 \ \wedge \ s > -0.5 \ \text{(strong recovery signal)} \\ 0.40 & \text{trend} > 0 \ \text{(mild recovery)} \\ 0.15 & \text{otherwise (slow decay — stay defensive)} \end{cases}$$

Why asymmetric?

- Entering crisis: $\alpha = 0.70$ means 70 % of the new signal is adopted immediately. We shift capital to PPO within 1–2 days.
- Exiting crisis: $\alpha = 0.15$ means we wait $\sim$6–7 days to confirm recovery before shifting back to SPY. Avoids whipsaw from dead-cat bounces.

Appendix A7: SPY Weight — C-Style Dynamic Map

We use only the C-style map (`dynamic_w` $= 1$, `static_w` $= 0$):

$$w_c(s) = \text{clamp}\Big(\underbrace{0.7}_{\text{base}} + \frac{s}{\hat{\sigma}_{\text{long}}},\ 0,\ 1\Big), \qquad \hat{\sigma}_{\text{long}} = \text{std}\big(s_{\text{buf}}[-260:]\big)$$

Intuition: when the macro score is positive and large relative to its historical variability, SPY gets high weight. When the score is negative, SPY weight drops toward zero—capital flows to PPO.

Crisis-adjusted SPY target:

$$w_{\text{spy}} = w_c \times (1 - c_t)$$

Floor constraint: PPO always gets at least 30 % gross.

$$w_{\text{spy}} \le 1 - \frac{g_{\min}}{0.95} = 1 - \frac{0.30}{0.95} \approx 0.684$$

Appendix A8: PPO Gross Exposure and Capital

Given the SPY weight, the remaining capital goes to PPO:

PPO gross exposure:

$$g_{\text{ppo}} = 0.95\,(1 - w_{\text{spy}})$$

PPO capital:

$$K_{\text{ppo}} = (1 - w_{\text{spy}}) \times \text{Equity}$$

Worked example:

$$s_{\text{macro}} = 0.4, \quad c_t = 0.1, \quad \hat{\sigma}_{\text{long}} = 0.25, \quad \text{Equity} = \$20\text{M}$$

Step	Computation
C-style base	$w_c = \text{clamp}(0.7 + 0.4/0.25,\ 0,\ 1) = \text{clamp}(2.3) = 1.0$
Crisis adjust	$w_{\text{spy}} = 1.0 \times (1 - 0.1) = 0.90$
Cap enforce	$w_{\text{spy}} = \min(0.90,\ 0.684) = 0.684$
PPO gross	$g_{\text{ppo}} = 0.95 \times (1 - 0.684) = 0.300$
PPO capital	$K_{\text{ppo}} = 0.316 \times \$20\text{M} = \$6.32\text{M}$

Appendix A9: Full Allocation Pipeline

8 Macro Indicators → Weighted Vote → Score Buffer → Trend, Vol, Vol Accel → Crisis Intensity c_t → $w_{\text{spy}},\ g_{\text{ppo}}$

Summary of key formulas:

$$
\begin{aligned}
s_{\text{macro}} &= \frac{\sum v_i \omega_i}{\sum \omega_i} && \text{(weighted vote of 8 indicators)} \\
c_t &= \alpha\, c_{\text{raw}} + (1-\alpha)\, c_{t-1} && \text{(asymmetric EMA, } \alpha \in \{0.15, 0.40, 0.60, 0.70\}\text{)} \\
w_{\text{spy}} &= \text{clamp}\big(0.7 + s/\hat{\sigma},\ 0,\ 1\big) \times (1 - c_t) && \text{(C-style} \times \text{crisis discount)} \\
g_{\text{ppo}} &= 0.95\,(1 - w_{\text{spy}}) && \text{(residual to PPO)}
\end{aligned}
$$

Appendix A10: Ramp-Up Period Allocation

During the first 2 years (2015-01-01 to 2017-01-01), allocation is **fixed** regardless of macro conditions:

$$w_{\text{spy}} = 0.80, \qquad g_{\text{ppo}} = 0.20$$

PPO capital during ramp-up:

$$K_{\text{ppo}} = \text{Equity} \times \frac{g_{\text{ppo}}}{\text{gross\_budget\_frac}} = \text{Equity} \times \frac{0.20}{0.95} \approx 0.211 \times \text{Equity}$$

Why this design?

- PPO's random policy creates unpredictable L/S positions.
- 80 % SPY provides a stable capital base, preventing margin calls.
- Even a total wipeout of the PPO book costs only ∼10 % of portfolio.
- Macro dynamics (trend, vol, crisis) are still tracked internally, so the allocator is "warm" when ramp-up ends.

Appendix A11: 15 Alpha Factors

k	Factor	Formula (sign = desired direction)
1	mom_5	$P_t/P_{t-5} - 1$
2	mom_20	$P_t/P_{t-20} - 1$
3	mom_60	$P_t/P_{t-60} - 1$
4	rev_2	$-(P_t/P_{t-2} - 1)$
5	rev_5	$-(P_t/P_{t-5} - 1)$
6	vol_10	$-\text{std}(r_{t-9:t})$
7	vol_20	$-\text{std}(r_{t-19:t})$
8	hlv_20	$-\text{mean}\big((H-L)/P\big)_{20}$
9	rsi_14	$-\text{RSI}(14)$
10	trend_20	$P_t/\text{MA}_{20} - 1$
11	gap_1	$P_t/P_{t-1} - 1$
12	skew_20	skewness of r_{20}
13	kurt_20	excess kurtosis of r_{20}
14	dvol_20	$-\text{std}(r_{<0})_{20}$
15	maxdd_60	max drawdown over 60 d

All factors are cross-sectionally winsorised (1 %) then z-scored. PPO outputs $w \in \mathbb{R}^{15}$; stock score $= Z_i^\top w$.

Appendix A12: Portfolio Construction & Beta Hedge

Scoring: $\text{score}_i = Z_i^\top w, \quad Z \in \mathbb{R}^{N\times 15}, \; w \in \mathbb{R}^{15}$ (PPO action).
Selection: Top-80 by score $\rightarrow$ long; Bottom-80 $\rightarrow$ short.
Weighting: Within each side, weight $\propto |\text{score}|$; rescaled so $\sum |h_L| = \frac{0.95}{2} K_{\text{ppo}}$ (same for short side).
Position bounds: $h_{\min} = 0.5\,\% \cdot K_{\text{ppo}}, \; h_{\max} = 15\,\% \cdot K_{\text{ppo}}$.
Beta neutralisation (up to 40 iterations):

1. Compute $\hat{\beta} = \sum_i h_i \beta_i / K_{\text{ppo}}$.
2. If $|\hat{\beta}| \le 0.10$, stop.
3. Find best (i, j) pair (one long, one short) whose $\beta_i - \beta_j$ can tilt $\hat{\beta}$ toward 0.
4. Shift Δ dollars from j to i subject to position bounds.
5. Repeat.

Appendix A13: PPO Reward Function

The agent receives a **PPO-isolated return** (SPY contribution stripped):

$$r_{\text{ppo}} = \frac{r_{\text{portfolio}} - w_{\text{spy}} \cdot r_{\text{SPY}}}{g_{\text{ppo}}}$$

Composite reward:

$$R = r_{\text{ppo}} - \underbrace{\lambda_\beta \max(0, |\beta| - 0.10)}_{\text{beta penalty}} - \underbrace{\lambda_f \cdot \text{fee\%}}_{\text{commission}} - \underbrace{\lambda_b \cdot \text{borrow\%}}_{\text{borrow cost}} - \underbrace{\eta \cdot \text{turnover\%}}_{\text{turnover}} - \underbrace{\lambda_\tau \cdot \text{tax\%}}_{\text{div WHT}}$$

Coefficient	Value
λ_β	2.0
λ_f	1.0
λ_b	1.0
η	0.05
λ_τ	1.0

Appendix A14: Transaction Cost Formulas

Slippage (`SpreadSlippageModel`):

$$\text{slip} = P \times 0.00005 \quad \text{per side (0.5 bps)}$$

Commission (`IBTieredFeeModel`):

$$\text{fee} = \max\big(\$0.35,\ \ \min(Q \times \$0.0035,\ \ Q \cdot P \cdot 0.005)\big)$$

Daily borrow charge:

$$\text{borrow}_t = \sum_{i:\, h_i<0} |h_i \cdot P_i| \ \times\ \frac{0.03}{252}$$

Dividend withholding tax:

$$\text{WHT}_j = D_j \times Q_j \times 0.30 \qquad (Q_j > 0 \text{ only})$$

Total execution cost per period:

$$C = \sum \text{slippage} + \sum \text{fee} + \sum \text{borrow}_t + \sum \text{WHT}_j$$